

Average Elasticity Estimation Report

Scope

This report estimates two objects for U.S. consumption categories using only local files in `Demand Elasticity`:

1. Primary target: share-weighted mean Marshallian own-price elasticity
$$\bar{\epsilon}^M = \sum_i s_i \epsilon_{ii}^M$$
2. Secondary target: IO-linked realized own-price elasticity approximation
$$\bar{\epsilon}^{GE} = \sum_i s_i \tilde{\epsilon}_{ii}$$

What Was Estimable

- **Estimable directly**: broad CEX category expenditures, CPI broad-category price indices, and BEA sector IO requirements.
- **Not fully identifiable**: a full structural GE/CGE model and exact paper-specific implementations (Jorgenson-Slesnick / Shojaeddini / Döpper / Inforum-LIFT) from local files alone.
- **Implemented closest defensible approximation**:
 - LA/AIDS-style broad demand system with adding-up and post-estimation homogeneity/symmetry projection.
 - IO-linked propagation using BEA domestic requirements matrix and sector-to-category mapping.

Category Definitions

See `results/category\_crosswalk.csv` for exact mappings and assumptions.

Marshallian Demand Estimation

Estimated share equations:

$$w_i = \alpha_i + \sum_j \gamma_{ij} \ln p_j + \beta_i \ln \left(\frac{x}{P} \right) + Z_i' \delta + u_i$$

with Stone index $\ln P = \sum_k w_k \ln p_k$, estimated by weighted equation-by-equation WLS (robustness: ridge regularization).

Elasticities at weighted mean shares:

$$\epsilon_{ij}^M = -\mathbf{1}(i=j) + \frac{\gamma_{ij}}{w_i} - \frac{\beta_i}{w_i} w_j$$

Preferred estimate:

- $\bar{\epsilon}^M = -0.4175$ (bootstrap 95% CI: [-2.3605, 1.1197])
- Identification quality (conservative rank): **weak**

Marshallian own elasticities:

category	share	marshallian_own_elasticity	contribution_share_times_elasticity	ci_lower_95	ci_upper_95
food_beverages	0.2241	-0.2245	-0.0503	-1.7495	1.0379
housing	0.4207	2.1388	0.8998	-2.2983	4.9660
apparel	0.0210	-10.5832	-0.2221	-27.9706	-2.6203
transportation	0.1444	-1.6668	-0.2407	-2.5021	-0.6020
medical_care	0.0961	-4.2581	-0.4093	-9.3999	-0.7212
recreation	0.0543	-6.3867	-0.3466	-18.7395	5.6163
education_communication	0.0129	-0.5619	-0.0072	-18.3702	13.8741
other_goods_services	0.0265	-1.5491	-0.0411	-15.1574	6.1040

IO-Linked Realized Elasticity Approximation

Construction:

1. Build sector requirement matrix $\backslash(L\backslash)$ from `CxC\_Domestic\_Sector.xlsx` (2017 sector table).
2. Recover direct-requirement analogue $\backslash(A = I - L^{\{-1\}}\backslash)$.
3. Propagate cost shocks via $\backslash((I-A')^{\{-1\}}\backslash)$.
4. Map sector price changes to consumption categories.
5. Map induced category price vector into quantity changes using estimated Marshallian matrix.
6. Define realized own elasticity for each category:

```
\[
\tilde{\backslash varepsilon}_{ii} = \frac{\backslash Delta \backslash ln Q_i}{\backslash Delta \backslash ln P_i}
\]
```

Preferred approximate estimate:

- $\backslash \bar{\backslash varepsilon}^{\{GE\}} = -1.8036 \backslash$ (IO-linked approximation, not a fully identified full-GE object)

Realized own elasticities:

category	share	realized_own_elasticity	contribution_share_times_realized_elasticity
food_beverages	0.2241	0.2237	0.0501
housing	0.4207	0.4665	0.1963
apparel	0.0210	-8.9529	-0.1879
transportation	0.1444	-1.6948	-0.2448
medical_care	0.0961	-5.1694	-0.4969
recreation	0.0543	-4.9939	-0.2710
education_communication	0.0129	-14.3672	-0.1847
other_goods_services	0.0265	-25.0776	-0.6647

Robustness Summary

scenario	weighted_mean_marshallian
baseline_8cat_regional	-0.4175
alternative_7cat_grouping	-1.4350
national_only_prices	-0.4623
regional_plus_rpp	-0.2711
regularized_ridge	-0.4203
exclude_housing	-1.2185
pce_inferred_shares_weighting	-3.1762

Limitations

- Public CEX geography and aggregate-category construction limit clean causal identification.
- Education/communication and other-goods/services mappings are approximate at this data granularity.
- Apparel lacks regional CPI in provided broad files; national fallback is used.
- GE object is an IO-linked propagation approximation, not an estimated structural CGE equilibrium.